# MSG No. 97.153 $I4'22'$ [ Type III, tetragonal ]

Table 1: Wyckoff site: `2a`, site symmetry: `4'22'`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ 0]$ | `[1,2,3,4,5,6,7,8]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[9,10,11,12,13,14,15,16]` |

Table 2: Wyckoff site: `2b`, site symmetry: `4'22'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ 0,\ \frac{1}{2}\right]$ | `[1,2,3,4,5,6,7,8]` |
| 2 | $\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | `[9,10,11,12,13,14,15,16]` |

Table 3: Wyckoff site: `4c`, site symmetry: `222.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ 0\right]$ | `[1,2,3,4]` |
| 2 | $\left[\frac{1}{2},\ 0,\ 0\right]$ | `[5,6,7,8]` |
| 3 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$ | `[9,10,11,12]` |
| 4 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[13,14,15,16]` |

Table 4: Wyckoff site: `4d`, site symmetry: `2.2'2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$ | `[1,4,15,16]` |
| 2 | $\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$ | `[2,3,13,14]` |
| 3 | $\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$ | `[5,6,10,11]` |
| 4 | $\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$ | `[7,8,9,12]` |

Table 5: Wyckoff site: `4e`, site symmetry: `4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ z]$ | `[1,4,5,6]` |
| 2 | $[0,\ 0,\ -z]$ | `[2,3,7,8]` |
| 3 | $\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | `[9,12,13,14]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | `[10,11,15,16]` |

Table 6: Wyckoff site: `8f`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{2},\ z\right]$ | `[1,4]` |
| 2 | $\left[0,\ \frac{1}{2},\ -z\right]$ | `[2,3]` |
| 3 | $\left[\frac{1}{2},\ 0,\ z\right]$ | `[5,6]` |

*continued ...*

Table 6

| No. | position | mapping |
|---|---|---|
| 4 | $\left[\frac{1}{2},\ 0,\ -z\right]$ | [7,8] |
| 5 | $\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$ | [9,12] |
| 6 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$ | [10,11] |
| 7 | $\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$ | [13,14] |
| 8 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$ | [15,16] |

Table 7: Wyckoff site: 8g, site symmetry: ..2'

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ x,\ 0]$ | [1,7] |
| 2 | $[x,\ -x,\ 0]$ | [2,6] |
| 3 | $[-x,\ x,\ 0]$ | [3,5] |
| 4 | $[-x,\ -x,\ 0]$ | [4,8] |
| 5 | $\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | [9,15] |
| 6 | $\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | [10,14] |
| 7 | $\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | [11,13] |
| 8 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | [12,16] |

Table 8: Wyckoff site: 8h, site symmetry: .2.

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ 0]$ | [1,2] |
| 2 | $[-x,\ 0,\ 0]$ | [3,4] |
| 3 | $[0,\ x,\ 0]$ | [5,7] |
| 4 | $[0,\ -x,\ 0]$ | [6,8] |
| 5 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | [9,10] |
| 6 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | [11,12] |
| 7 | $\left[\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$ | [13,15] |
| 8 | $\left[\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$ | [14,16] |

Table 9: Wyckoff site: 8i, site symmetry: .2.

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ 0,\ \frac{1}{2}\right]$ | [1,2] |
| 2 | $\left[-x,\ 0,\ \frac{1}{2}\right]$ | [3,4] |
| 3 | $\left[0,\ x,\ \frac{1}{2}\right]$ | [5,7] |
| 4 | $\left[0,\ -x,\ \frac{1}{2}\right]$ | [6,8] |
| 5 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | [9,10] |
| 6 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$ | [11,12] |
| 7 | $\left[\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$ | [13,15] |
| 8 | $\left[\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$ | [14,16] |

Table 10: Wyckoff site: 8j, site symmetry: ..2'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\, x+\frac{1}{2},\, \frac{1}{4}\right]$ | [1,15] |
| 2 | $\left[x,\, \frac{1}{2}-x,\, \frac{3}{4}\right]$ | [2,14] |
| 3 | $\left[-x,\, x+\frac{1}{2},\, \frac{3}{4}\right]$ | [3,13] |
| 4 | $\left[-x,\, \frac{1}{2}-x,\, \frac{1}{4}\right]$ | [4,16] |
| 5 | $\left[\frac{1}{2}-x,\, x,\, \frac{1}{4}\right]$ | [5,11] |
| 6 | $\left[x+\frac{1}{2},\, -x,\, \frac{1}{4}\right]$ | [6,10] |
| 7 | $\left[x+\frac{1}{2},\, x,\, \frac{3}{4}\right]$ | [7,9] |
| 8 | $\left[\frac{1}{2}-x,\, -x,\, \frac{3}{4}\right]$ | [8,12] |

Table 11: Wyckoff site: 16k, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, y,\, z]$ | [1] |
| 2 | $[x,\, -y,\, -z]$ | [2] |
| 3 | $[-x,\, y,\, -z]$ | [3] |
| 4 | $[-x,\, -y,\, z]$ | [4] |
| 5 | $[-y,\, x,\, z]$ | [5] |
| 6 | $[y,\, -x,\, z]$ | [6] |
| 7 | $[y,\, x,\, -z]$ | [7] |
| 8 | $[-y,\, -x,\, -z]$ | [8] |
| 9 | $\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$ | [9] |
| 10 | $\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$ | [10] |
| 11 | $\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [11] |
| 12 | $\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$ | [12] |
| 13 | $\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$ | [13] |
| 14 | $\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$ | [14] |
| 15 | $\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$ | [15] |
| 16 | $\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$ | [16] |